BIS RAG / Vector DB Adversarial Test Benchmark

Procurement standards retrieval, edition control, scope matching, and hallucination diagnostics

Purpose

This benchmark is designed to determine whether an e-Procurement RAG system is failing because of stale or incorrect vector-database content, weak retrieval/reranking, missing standards metadata, incorrect model selection, or lack of post-generation validation.

Key principle: evaluate the retrieved evidence as well as the final tender text. A correct final answer can hide poor retrieval, while a wrong final answer can originate from several different layers.

8-Test Benchmark

Test	Product / Prompt	Expected behavior	Primary diagnosis
1	TMT steel bars, Fe 500D	Retrieve IS 1786:2008 and treat it as the applicable product standard for the specified reinforcement steel.	Basic correct retrieval / product-to-standard matching
2	TMT steel bars - explicitly ask for IS 1786:1985	Identify the old edition and warn or substitute the applicable current edition rather than silently presenting the legacy edition as current.	Edition / deprecation handling
3	16 A, 250 V plug and socket-outlet	Do not treat IS 1293:2005 as current; identify the current applicable edition and flag the superseded 2005 edition.	Superseded-standard detection
4	1.1 kV XLPE insulated power cable	Retrieve IS 7098 Part 1:2025 for the 1,100 V class, subject to exact cable construction and procurement requirements.	Part / voltage-range matching
5	1.1 kV XLPE cable with Part 2 seeded strongly in the DB	Reject IS 7098 Part 2 when its voltage scope does not cover the requested 1.1 kV cable; choose the correct part based on scope, not semantic similarity alone.	Wrong-part detection despite correct IS family
6	Mandatory-QCO cement specification with no IS number in the prompt	Retrieve the applicable product standard and mandatory conformity/QCO information from the knowledge base without requiring the user to supply the IS code.	Missing-standard retrieval
7	Crystalline-silicon solar PV module with electrical safety requirements	Retrieve the primary PV standard plus applicable allied safety standards when the prompt asks for both product qualification and safety compliance.	Multi-standard / allied-standard retrieval
8	11 kV, 1000 kVA outdoor distribution transformer	Retrieve the applicable transformer product standard for the described equipment and ratings.	Product-to-standard semantic matching

High-Value Diagnostic Test: 1.1 kV XLPE Cable

Use this as a controlled adversarial test. Seed the corpus with the following scope-separated records: **IS 7098 Part 1:2025 - up to and including 1,100 V**; **IS 7098 Part 2 - higher-voltage XLPE cable range**; and **IS 694 - PVC insulated cable scope**. Then ask only: *"Applicable BIS standard for 1.1 kV XLPE cable?"*

Expected result: the system should select **IS 7098 Part 1:2025**, and it should be able to explain why Part 2 and IS 694 do not fit the requested construction/voltage scope. This is stronger than a simple semantic similarity test because the correct answer depends on structured scope metadata.

What to Log for Every Test

Query; top-K retrieved chunks; source document IDs; retrieved IS numbers; edition/year; part/sub-part; stated voltage/application scope; similarity score; reranker score; final model answer; validation result; deprecation/currentness flag.

Failure-Layer Interpretation

Observed result	Likely issue
Wrong Top-K evidence	Vector DB content, embeddings, chunking, metadata, or retrieval filter problem.
Correct evidence retrieved, wrong selection	Reranker, prompt logic, or model selection problem.
Correct standard retrieved, wrong final answer	Generation/reasoning or grounding problem.
Correct standard, no deprecation warning	Missing currentness/edition metadata or validation layer.
Correct IS family, wrong Part	Missing structured part/scope metadata or insufficient rule-based validation.
Correct answer but unsupported alternative appears	Hallucination / weak grounding / failure to constrain output to retrieved evidence.

Suggested Validation Rule

Product + construction + rating/voltage -> candidate IS family -> part/section scope check -> edition/currentness check -> QCO applicability check -> final generation.

Do not allow semantic similarity alone to decide between standards that share the same family number but have different parts, voltage ranges, construction types, or editions.

Example Controlled Test Record

Field	Test value
Query	Applicable BIS standard for 1.1 kV XLPE cable?
Top-K expected	IS 7098 Part 1:2025; IS 7098 Part 2; IS 694
Correct selection	IS 7098 Part 1:2025
Reject	Part 2 if its voltage range does not cover the requested 1.1 kV class; IS 694 when the cable construction is XLPE rather than PVC
Pass condition	Final answer names the correct part and cites the supporting retrieved evidence
Fail condition	Returns Part 2 merely because it has the same IS family number or because of higher similarity score

Benchmark Outcome Goal

The benchmark should distinguish **retrieval correctness** from **generation correctness**. A production procurement system should not rely on the LLM alone to detect superseded standards, select the correct sub-part, or infer scope from a vector similarity score. Regulatory metadata and deterministic validation should participate in the decision.

Note: BIS references and currentness should be re-verified against the applicable BIS publication/QCO before production use, because standards and regulatory status can change.